MATH 130: 10/22 WORKSHEET

Last time we looked at measures of the *center* of a data set. Now we turn to measuring the *spread* of a data set.

Minimum, maximum, and range.
Consider a data set

$$x_1, x_2, \ldots, x_n.$$

The *minimum* is the smallest value in the data set and the maximum is the largest value in the data set. That is, the minimum is the value x_m so that $x_m \leq x_i$ for every index i. And the maximum is the value x_M so that $x_M \geq x_i$ for every index i. The *range* of a data set is the maximum minus the minimum. The range is the total length of the interval in which values from the data set lie.

Quartiles.
Quartiles are measures of where to break up a data set into four equally sized chunks. The *first quartile* is where 25% of values are smaller, the *second quartile* is another name for the median, and the *third quartile* is where 75% of values are smaller.

Quantiles.
Generalizing the median and quartiles, you can split a data set into any number of pieces. Other common choices are *quintiles* (five pieces) and *deciles* (ten pieces).